National Telecommunication Institute

Summer Training Project

Project: Retail Customer Purchase Behavior & Sales Performance

Group Code: S26-B3-Data A-G7

Project Number: 1

Project Members

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Contents

1.Introduction

The company has collected two years of online retail transaction data (2009–2011). However, despite having a large amount of sales data, it lacks clear insights into its business performance. The company does not fully understand which products generate the highest revenue, who its most valuable customers are, or how sales vary across different countries and times.

Without analyzing this data, important business decisions—such as inventory planning, marketing campaigns, and customer relationship strategies—are often based on assumptions rather than evidence. This can lead to missed sales opportunities, inefficient resource allocation, and reduced business performance.

2.Business Scenario

An online retail company aims to improve its business performance by making data-driven decisions. The company has accumulated two years of transactional sales data but lacks a clear understanding of customer purchasing behavior, product performance, and sales trends across different countries and time periods.

Management wants to identify the products that contribute the most to revenue, recognize their most valuable customers, understand seasonal demand patterns, and evaluate sales performance across international markets. These insights will support strategic decisions related to inventory management, marketing campaigns, customer retention, and future business planning.

As a data analyst, the objective of this project is to transform raw sales data into meaningful business insights through data exploration, cleaning, visualization, and analytical techniques. The outcome is intended to help management optimize operations and improve profitability for the next business quarter.

3.Dataset Description

This project uses the **Online Retail II** dataset obtained from the **UCI Machine Learning Repository**. The dataset contains transactional records from a UK-based online retail company that primarily sells unique gift items to customers across multiple countries.

The dataset is provided as an Excel workbook containing **two worksheets**, each representing a different period of sales transactions:

Worksheet	Time Period
Year 2009–2010	December 1, 2009 – December 9, 2010
Year 2010–2011	December 1, 2010 – December 9, 2011

Since both worksheets have the same structure and attributes, they were merged into a single dataset before the analysis. Combining the two worksheets provides a complete view of approximately two years (24 months) of sales activity, allowing more accurate trend analysis, customer segmentation, and business insights.

Each row in the dataset represents a single product within a customer invoice. Therefore, one invoice may contain multiple rows if multiple products were purchased in the same transaction.

The dataset includes the following attributes:

Attribute	Description
Invoice	Unique invoice number for each transaction. Cancelled invoices usually begin with the letter **"C"**.
StockCode	Unique identifier assigned to each product.
Description	Product name or description.
Quantity	Number of units purchased. Negative values usually indicate returned or cancelled items.
InvoiceDate	Date and time when the transaction was recorded.
Price	Unit price of the purchased product.
Customer ID	Unique identifier assigned to each customer. Some records have missing customer IDs.
Country	Country where the customer is located.

The dataset is suitable for performing sales analysis, customer behavior analysis, revenue analysis, customer segmentation using the RFM model, and identifying sales trends across different products, countries, and time periods.

4.Solution Steps

Python

Load data

2 Data Frames (each sheet in a DataFrame)

Exploration

1. Show top 10 rows of data
2. Shape
3. Columns name
4. Types of columns
5. Number of total transactions
6. Number of unique customers
7. Number of unique products
8. Number of countries
9. Date range
10. Name of unique countries
11. Description for new/working columns
12. Top 20 rows where Price = 0
13. Show new/working columns
14. Missing values
15. Duplicates
16. Outliers
17. Inconsistence (Negative Quantity and Price)
18. Identify canceled transactions

Cleaning

1. Remove duplicates
2. Fill Description by "Unknown"
3. Fill Description by Description has same StockCode
4. Handle CustomerID
5. Fill Price = 0 by price of the same StockCode and Description
6. Drop Price with null
7. Drop inconsistent Quantity
8. Drop inconsistent Price
9. Handle positive quantity in canceled transactions
10. Drop inconsistent Price
11. Standardizing fields
12. Separate DataFrame from canceled transactions

Integration

four Data Frame Integration Give the two Data Frames:

1. Uncanceled Transactions (df)
2. Canceled Transactions (df_canceled)

CSV

Convert two Data Frames to two CSV files:

1.NTI_Cleaned_Final.csv

2.NTI_Cancelled_Clean.csv

Check

-Check all data in NTI_Cleaned_Final.csv

1. No zero price
2. No negative values
3. No null

4. No duplicates

5. No canceled transaction

-Check all data in NTI_Canceled_Cleaned_Final.csv

1. No zero price

2. No positive values

3. No null

4. No duplicates

5. No uncanceled transaction

SQL

Business and Diagnostic Analysis Questions

Setup in SQL server

First, make Database with any name and make 2 tables cleaned and canceled

And this is the valid design that we used

Column Name	Data Type
Invoice	nvarchar(50)
StockCode	nvarchar(50)
Description	nvarchar(150)
Quantity	int
InvoiceDate	datetime2(7)
Price	decimal(10, 5)
Customer_ID	float
Country	nvarchar(50)

Product Analysis

Q1. What are the top 10 products by revenue?

Purpose

Identify the products that generate the highest total revenue to understand which items contribute most to overall sales.

Expected Insight

- Top-selling products by revenue and not cancelled.
- Useful for inventory planning and marketing campaigns.

Q2. Which products generate the least revenue?

Purpose

Identify products with poor sales performance.

Expected Insight

- Low-performing products.
- Candidates for promotions or discontinuation.

Q3. Which products generate high revenue despite low sales quantity?

Purpose

Find premium or expensive products that earn high revenue even with low sales volume.

Expected Insight

- High-value products.
- Luxury or premium items.

Q4. Which products have high sales quantity but low revenue?

Purpose

Identify products sold in large quantities but generating relatively low revenue.

Expected Insight

- Low-priced products.
- Frequently purchased items.

Q5. Which products are canceled most frequently and what percentage do they represent?

Purpose

Analyze product cancellations to identify problematic items.

Expected Insight

- Frequently canceled products.
- Possible issues with quality, availability, or customer satisfaction.

Q6. Which pairs of products are most purchased together?

Purpose

Discover products frequently bought together.

Expected Insight

- Product bundles.

- Cross-selling opportunities.
- Recommendation systems.

Customer Analysis

Q7. Which customers order most frequently, and what is their average order value?

Purpose

Identify loyal customers and evaluate their purchasing behavior.

Expected Insight

- Most active customers.
- Average spending per order.

Q8. When did each customer place their last order?

Purpose

Determine the most recent purchase date for every customer.

Expected Insight

- Active customers.
- Inactive customers.
- Customer retention analysis.

Q9. Segment customers using RFM Analysis (Recency, Frequency, Monetary)?

Purpose

Classify customers into business segments based on purchase behavior.

Customer Segments

- High-Value Customers
- At-Risk Customers
- New Customers

Expected Insight

- Identify loyal customers.

- Detect customers likely to churn.
- Recognize newly acquired customers.

Time Analysis

Q10. How does monthly revenue change over the two years?

Purpose

Analyze revenue trends over time.

Expected Insight

- Monthly sales trend.
- Seasonal demand.
- Growth or decline in revenue.
-

Q11. Which day of the week generates the highest revenue?

Purpose

Identify the best-performing weekday.

Expected Insight

- Peak shopping days.
- Better scheduling for promotions and staffing

Country Analysis

Q12. Which countries generate the most revenue, and how concentrated is it?

Purpose

Determine revenue contribution by country.

Expected Insight

- Top revenue-generating countries.
- Revenue concentration by market.

Example Finding

The United Kingdom contributes approximately 86%of the total revenue.

Q13. Which country has the highest average quantity per order ?

Purpose

Compare purchasing behavior across countries.

Expected Insight

- Countries with larger average order sizes.
- Difference in market demand.

Revenue Analysis

Q14. What is the net revenue after removing discounts?

Purpose

Calculate the actual revenue after excluding discount transactions

Expected Insight

- Gross Revenue
- Total Discounts
- Net Revenue

Excel

Get Answers for All Questions (#14) Above and make dashboards

Visualization (Power PI)

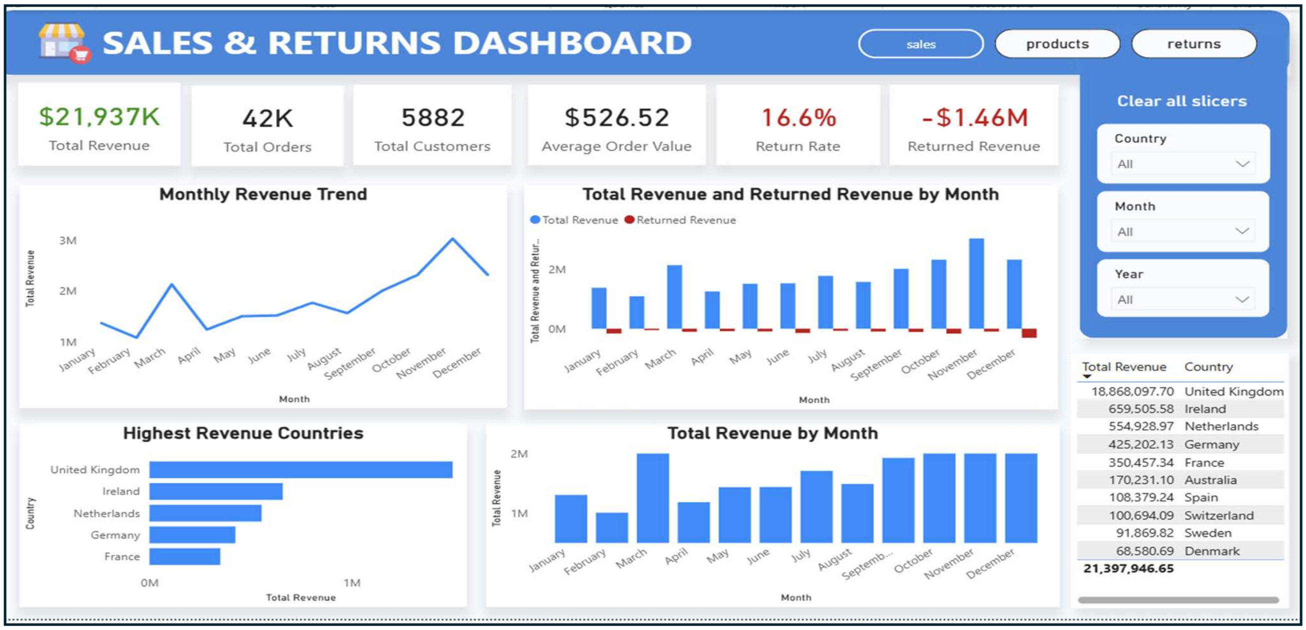
1-Import Data

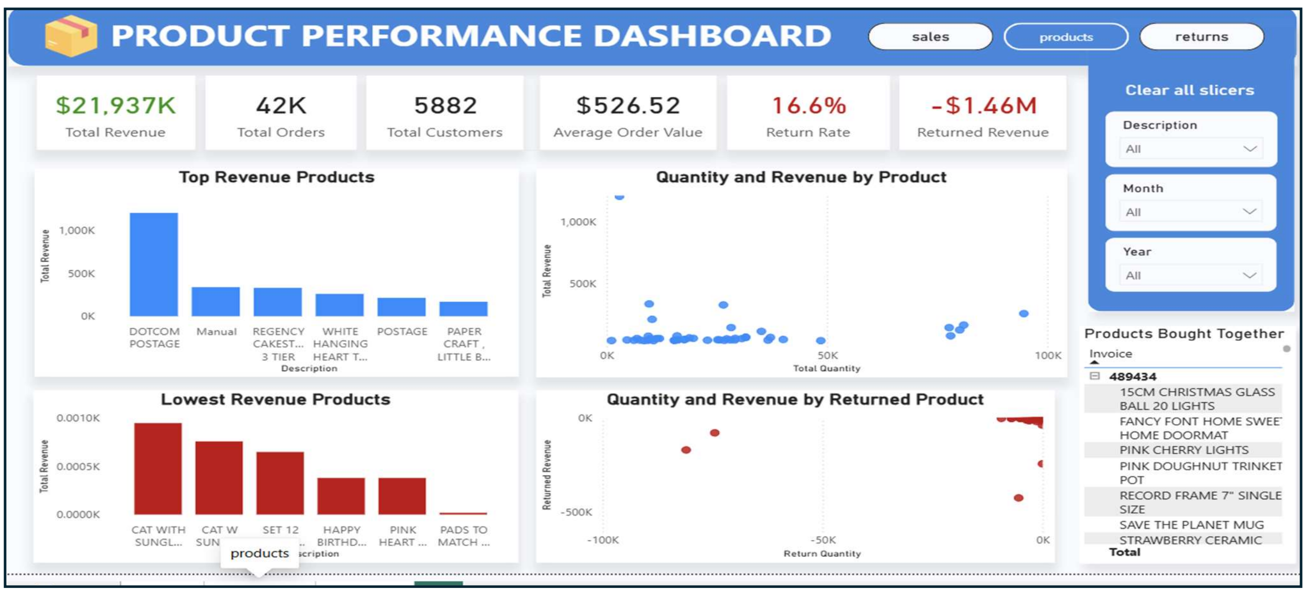
2-Data Modeling

3-Create Measures

4-Create Charts and tables

5-Create Interactive dashboard with 3 Pages (sales / products / returns)


SALES & RETURNS DASHBOARD
sales
products
returns
$21,937K
Total Revenue
42K
Total Orders
5882
Total Customers
$526.52
Average Order Value
16.6%
Return Rate
-$1.46M
Returned Revenue
Clear all slicers
Country
All
Month
All
Year
All
Monthly Revenue Trend
Total Revenue
3M
2M
1M
January February March April May June July August September October November December
Month
Total Revenue and Returned Revenue by Month
Total Revenue
Returned Revenue
Total Revenue and Retur...
2M
0M
Highest Revenue Countries
Country
United Kingdom
Ireland
Netherlands
Germany
France
0M
1M
Total Revenue
Total Revenue by Month
2M
1M
Total Revenue
Total Revenue Country
18,868,097.70 United Kingdom
659,505.58 Ireland
554,928.97 Netherlands
425,202.13 Germany
350,457.34 France
170,231.10 Australia
108,379.24 Spain
100,694.09 Switzerland
91,869.82 Sweden
68,580.69 Denmark
21,397,946.65


PRODUCT PERFORMANCE DASHBOARD
sales
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$21,937K
Total Revenue
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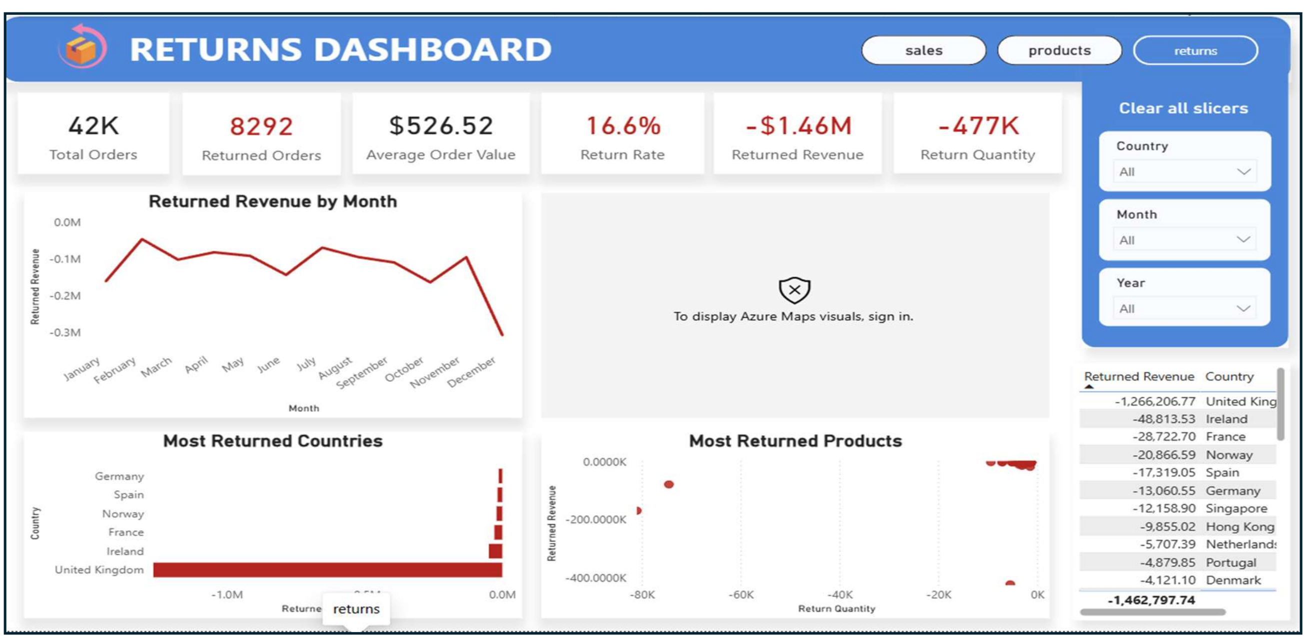
Returned Revenue
Clear all slicers
Description
All
Month
All
Year
All
Top Revenue Products
Total Revenue
1,000K
500K
0K
DOTCOM POSTAGE
Manual
REGENCY CAKEST... 3 TIER
WHITE HANGING HEART T...
POSTAGE
PAPER CRAFT , LITTLE B...
Description
Quantity and Revenue by Product
1,000K
500K
0K
50K
100K
Total Quantity
Products Bought Together
Invoice
489434
15CM CHRISTMAS GLASS BALL 20 LIGHTS
FANCY FONT HOME SWEET HOME DOORMAT
PINK CHERRY LIGHTS
PINK DOUGHNUT TRINKET POT
RECORD FRAME 7" SINGLE SIZE
SAVE THE PLANET MUG
STRAWBERRY CERAMIC
Total
Lowest Revenue Products
0.0010K
0.0005K
0.0000K
CAT WITH SUNGL...
SET 12
HAPPY BIRTHD...
PINK HEART ...
PADS TO MATCH ...
products
Quantity and Revenue by Returned Product
Returned Revenue
0K
-500K
-100K
-50K
0K
Return Quantity


RETURNS DASHBOARD
sales
products
returns
42K
Total Orders
8292
Returned Orders
$526.52
Average Order Value
16.6%
Return Rate
-$1.46M
Returned Revenue
-477K
Return Quantity
Clear all slicers
Country
All
Month
All
Year
All
Returned Revenue by Month
Returned Revenue
0.0M
-0.1M
-0.2M
-0.3M
January February March April May June July August September October November December
Month
To display Azure Maps visuals, sign in.
Returned Revenue Country
-1,266,206.77 United King
-48,813.53 Ireland
-28,722.70 France
-20,866.59 Norway
-17,319.05 Spain
-13,060.55 Germany
-12,158.90 Singapore
-9,855.02 Hong Kong
-5,707.39 Netherlands
-4,879.85 Portugal
-4,121.10 Denmark
-1,462,797.74
Most Returned Countries
Country
Germany
Spain
Norway
France
Ireland
United Kingdom
-1.0M
0.0M
returns
Most Returned Products
Returned Revenue
0.0000K
-200.0000K
-400.0000K
-80K
-60K
-40K
-20K
0K
Return Quantity